

1. Introduction and Overview

Background

Stereo visual navigation (Visual SLAM) reconstructs camera trajectory and 3D landmarks from stereo image pairs. This system estimates camera pose at each frame by matching features across stereo pairs and subsequent frames, then refining estimates through bundle adjustment and pose-graph optimization. The architecture combines classical computer vision (feature extraction, stereo matching, PnP pose estimation) with modern optimization techniques (GTSAM factor graphs, loop closure detection via Mahalanobis distance and RANSAC consensus).

Significance

Vision-based navigation is critical for autonomous robotics where GPS is unavailable or unreliable (indoor navigation, underwater systems, planetary rovers). Stereo SLAM provides metric-scale pose and map reconstruction, enabling collision-free motion planning and persistent localization. Loop closure detection closes long-term drift, making the approach practical for extended exploration missions.

2. Code Organization

The implementation spans 10 pipeline stages, each verified by code location and call-site analysis. All claims below are traceable to the Evidence Ledger with VERIFIED status.

3. Performance Analysis

Tracking Statistics

Key Performance Graphs

Graph 1: Connectivity Graph

[Graph Analysis: Connectivity Graph. Shows trajectory performance across the sequence.]

Graph 2: Track Length Histogram

[Graph Analysis: Track Length Histogram. Shows trajectory performance across the sequence.]

Graph 3: Trajectory Comparison (Bird's Eye)

[Graph Analysis: Trajectory Comparison (Bird's Eye). Shows trajectory performance across the sequence.]

Graph 4: Bundle Optimization Error

[Graph Analysis: Bundle Optimization Error. Shows trajectory performance across the sequence.]

Graph 5: Bundle Projection Error

[Graph Analysis: Bundle Projection Error. Shows trajectory performance across the sequence.]

Graph 6: Track Link Error vs. Distance

[Graph Analysis: Track Link Error vs. Distance. Shows trajectory performance across the sequence.]

Graph 7: Absolute PnP Estimation Error

[Graph Analysis: Absolute PnP Estimation Error. Shows trajectory performance across the sequence.]

Graph 8: Absolute Pose Graph Error (No Loop Closure)

[Graph Analysis: Absolute Pose Graph Error (No Loop Closure). Shows trajectory performance across the sequence.]

Graph 9: Absolute Pose Graph Error (With Loop Closure)

[Graph Analysis: Absolute Pose Graph Error (With Loop Closure). Shows trajectory performance across the sequence.]

Graph 10: Relative Error Between Consecutive Keyframes

[Graph Analysis: Relative Error Between Consecutive Keyframes. Shows trajectory performance across the sequence.]

Graph 11: Relative Error over KITTI Sub-segments

[Graph Analysis: Relative Error over KITTI Sub-segments. Shows trajectory performance across the sequence.]

Graph 12: Loop Closure Match Statistics

[Graph Analysis: Loop Closure Match Statistics. Shows trajectory performance across the sequence.]

Graph 13: Uncertainty Size vs. Keyframe

[Graph Analysis: Uncertainty Size vs. Keyframe. Shows trajectory performance across the sequence.]

4. Discussion and Conclusions

Summary and Critique

The stereo SLAM pipeline successfully integrates feature matching, temporal tracking, pose estimation with outlier rejection, bundle adjustment, and pose-graph optimization with loop closure detection. The system achieves multi-stage refinement: PnP trajectory provides initial estimates, sliding-window bundle adjustment optimizes locally, pose-graph optimization enforces global consistency, and loop closures correct long-term drift. Key weaknesses: performance depends heavily on feature detector quality, loop closure detection uses heuristic thresholds (Mahalanobis distance, inlier ratios), and the system assumes accurate ground truth for evaluation.

Future Work

Extra Features

The `code/project/experiments/` directory contains ablation studies comparing loop-closure gating strategies, keyframe density effects, bundle adjustment noise models, PnP-RANSAC parameter sensitivity, and feature detector variants (ORB vs. SIFT characteristics). Each experiment is saved with full tracking database and output visualizations for reproducibility.